



VIAVI Solutions

AEA Connect:

GPS Simulation in Aviation

September 2025

Mark Anderson, Applications Engineering

AGENDA

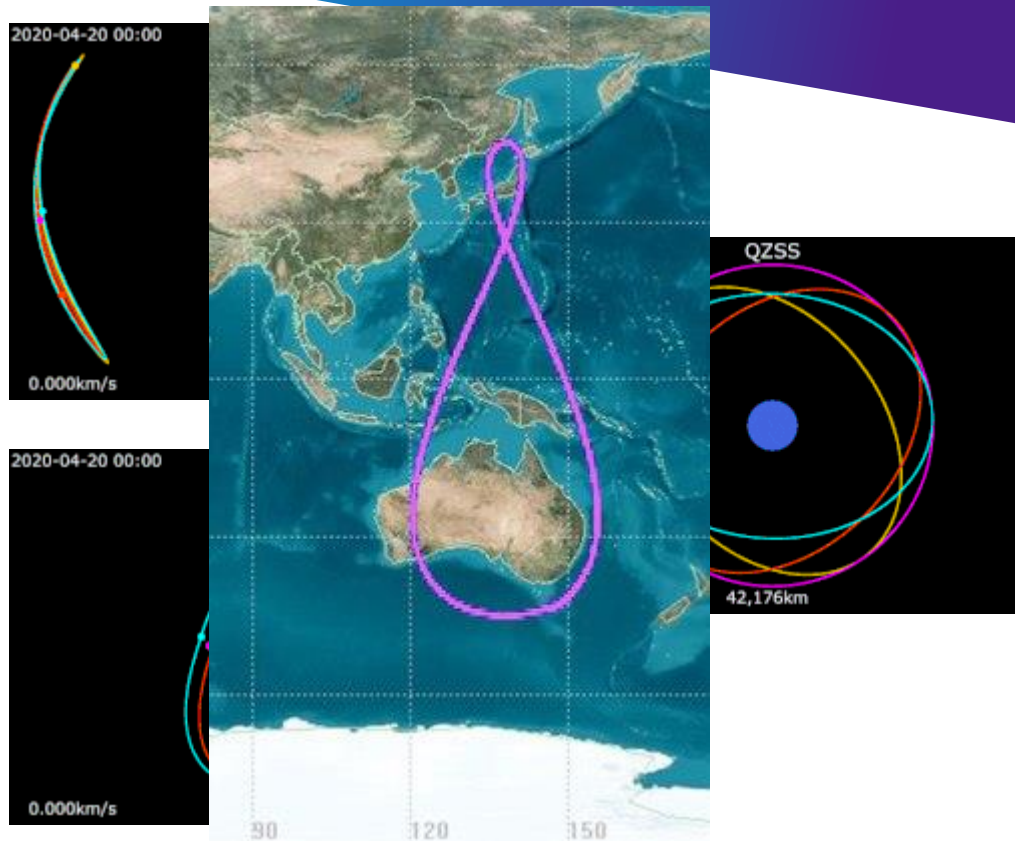
- GPS overview
- Almanac data
- Simulation Types
- Benefits of GPS Simulation
- Use Cases
- VIAVI GPS Solutions

GPS Overview

- Who does what now?

Constellation Breakdown:

- GPS (USA) : 31
- GLONASS (Russia): 24
- Galileo (European Union): 25
- BeiDou/BDS (China): 44
- NavIC (India): 7
- QZSS (Japan): 4



GPS Overview

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Orbits:

- [LEO \(~1200 miles\)](#)
- [MEO \(~12550 miles\)](#)
- [GEO \(22,236 miles\)](#)

GPS Satellites operate in Medium Earth Orbit while most other space equipment (Starlink: 7,875) operate in Low Earth Orbit

Terminology

SV

SBAS

WAAS

RNAV

LNAV

VNAV

LPV



GPS Overview

Terminology

GPS and AVIATION RELATED ACRONYMS

SV



SBAS

WAAS

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Space Vehicles: Individual satellites that make up the GPS constellations



GPS Overview

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SBAS



WAAS

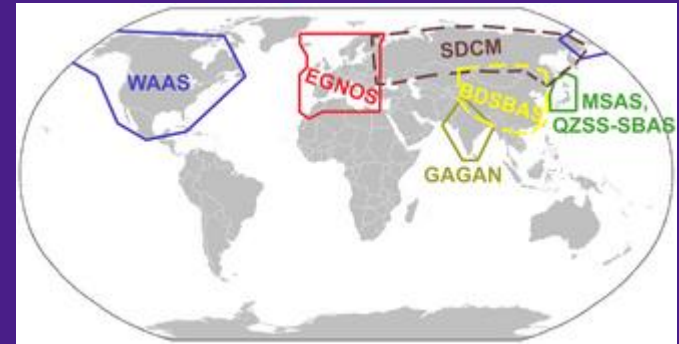
RNAV

LNAV

VNAV

LPV

Satellite Based Augmentation System:
A general term for systems like WAAS



GPS Overview

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GPS and AVIATION RELATED ACRONYMS

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SBAS

WAAS



RNAV

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Wide Area Augmentation System:
Improves accuracy, integrity, availability of GPS
(Geostationary SVs)

Oh, by the way:
GPS Accuracy without WAAS – 30 meters
GPS Accuracy with WAAS – 3 meters

GPS Overview

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SBAS

WAAS

RNAV

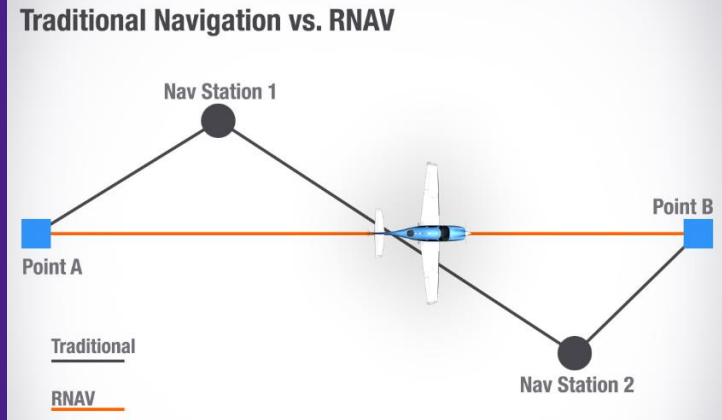
LNAV

VNAV

LPV



Area Navigation: Enables aircraft to navigate directly from one point to another, rather than being restricted to routes defined by ground-based navigation beacons.



GPS Overview

Terminology

GPS and AVIATION RELATED ACRONYMS

SV

SBAS

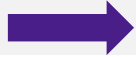
WAAS

RNAV

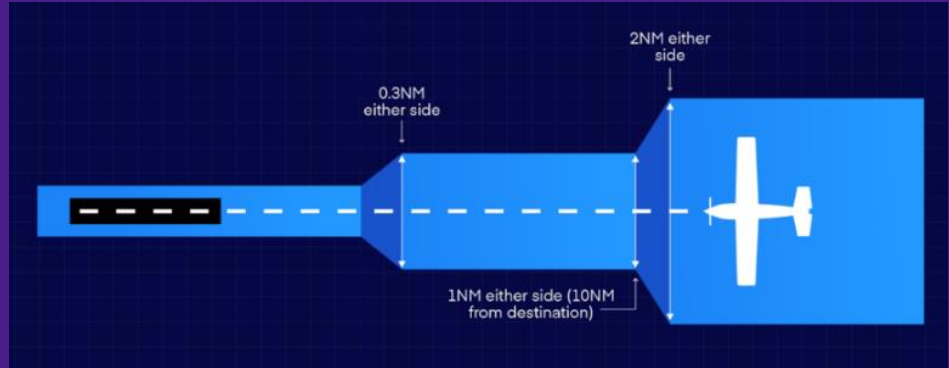
LNAV

VNAV

LPV



Lateral Navigation: Guides aircraft along a horizontal path (think Localizer)



GPS Overview

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SV

SBAS

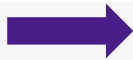
WAAS

RNAV

LNAV

VNAV

LPV



Vertical Navigation: Guides aircraft along a vertical path (think Glideslope). Combines with LNAV for approaches.

Lateral Navigation and Vertical
Navigation (LNAV/VNAV)



GPS Overview

Terminology

GPS and AVIATION RELATED ACRONYMS

SV

SBAS

WAAS

RNAV

LNAV

VNAV

LPV



Localizer Performance with Vertical Guidance:
GPS Instrument approach leveraging the accuracy
of WAAS

Localizer Performance with
Vertical Guidance (LPV)



GPS Overview

Spoofing – Is it all bad?

GPS Overview

Spoofing – Is it all bad?

- Yes.



GPS Overview

Spoofing – Is it all bad?

- Yes. GPS Spoofing is a term that implies a malicious attempt to trick a GPS receiver into believing a false location. Also see GPS Jamming.

- GPS Simulation? ... Good! ★

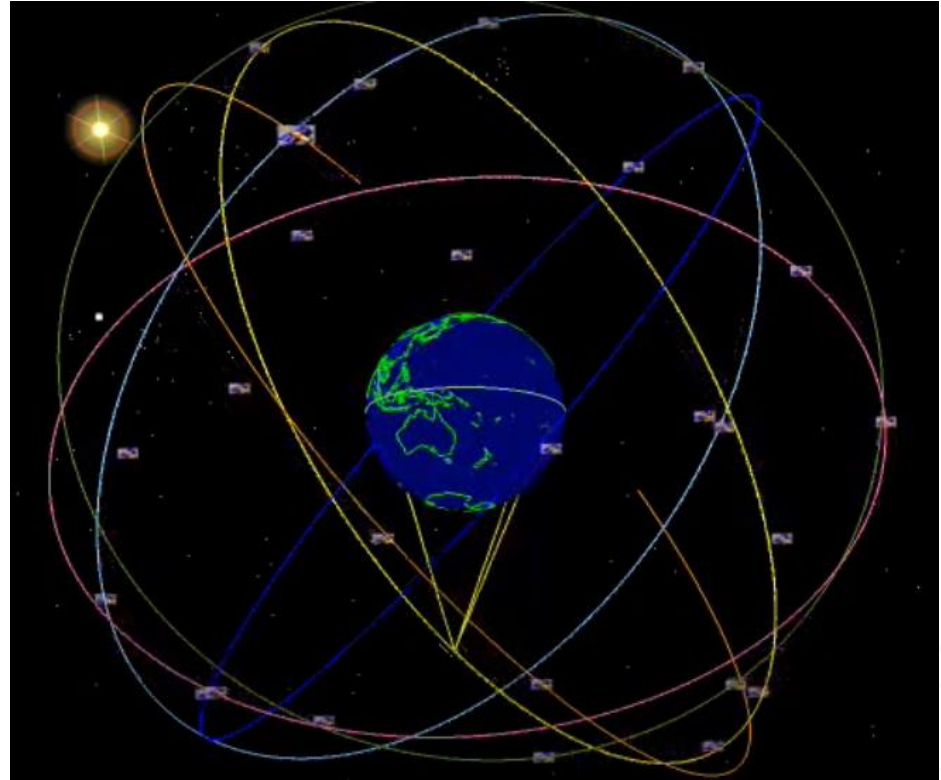
A testing method that uses a controlled environment to mimic real GPS signals for evaluating GPS receiver performance, security, accuracy, and sensitivity.

Almanac Data

What is it and why do I need it?

A reduced set of data that provides azimuth and elevation data for all of the Space Vehicles in the GPS constellation.

An almanac file provides spatial location information at a given point in time that simulators use as a baseline. Locations are then extrapolated, however, the longer the timeframe, the lower the accuracy, the less likely you are to obtain receiver GPS lock.



Almanac Data

What is it and why do I need it?

Some simulators require an Almanac file while others require a RINEX file.

Almanac data provided weekly by US Coast Guard:

```
***** Week 323 almanac for PRN-01 *****
ID:                01
Health:            000
Eccentricity:      0.5660057068E-003
Time of Applicability(s): 319488.0000
Orbital Inclination(rad): 0.9592137662
Rate of Right Ascen(r/s): -0.7954617056E-008
SQRT(A) (m 1/2):   5153.671875
Right Ascen at Week(rad): 0.1491287361E+001
Argument of Perigee(rad): 0.213221831
Mean Anom(rad):    -0.2524453008E+001
Af0(s):            0.2946853638E-003
Af1(s/s):          0.1091393642E-010
week:              32
```

RINEX data provided daily/hourly by NASA:

```
1 25 5 5 0 0 0.0 0.248158350587D-03 0.140971678775D-10
0.000000000000D+00 0.239000000000D+03 0.390000000000D+02
0.448697261471D-08 -0.242292608940D+00 0.211782753468D-05
0.515963882208D-03 0.725127756596D-05 0.515372778511D+04
0.864000000000D+05 -0.186264514923D-08 0.224562023971D+01-
0.279396772385D-07 0.959505714281D+00 0.240218750000D+03
0.144979004260D+00-0.809712299196D-08 0.101075638777D-09
0.100000000000D+01 0.236500000000D+04 0.000000000000D+00
0.200000000000D+01 0.000000000000D+00-0.931322600000D-08
0.751000000000D+03 0.842100000000D+05 0.400000000000D+01
0.000000000000D+00 0.000000000000D+00
```

RINEX: Receiver Independent Exchange Format

Simulation Types

Static vs. Dynamic vs. Trajectory

Static Waypoints – probably >80% of use cases

- Simulates one non-moving location's Latitude / Longitude / Altitude

Dynamic Routes

- A series of a number of waypoints that make up a flight / travel path

Trajectory Path (can include Record / Playback capability)

- Similar to Dynamic but uses high throughput NMEA data as the input. NMEA data is a standard protocol for information transfer. The commands look similar to this:

```
$GPGGA,123519,4807.038,N,01131.000,E,1,08,0.9,545.4,M,46.9,M,,*47
```

Benefits of GPS Simulation

When and Why?

Simulate numerous locations in and around your hangars

Verification of system installations

Efficient ADS-B testing

Train technicians on new avionics installs

Train pilots on upgraded avionics without flying the aircraft

Testing XM Weather systems

Support for SAFO 24002

Benefits of GPS Simulation

GPS Related Aircraft Systems

GPS technology now touches a lot of avionics-related equipment:

- Transponder w/ ADS-B
- Autopilot
- Traffic Collision Avoidance Systems (TCAS)
- Datalink
- Flight Director
- Synthetic Vision Systems (SVS)
- Terrain Awareness and Warning System (TAWS)
- Weather Systems

Consider the possibilities!

Use Cases

Anywhere GPS Solutions are needed

Use a static location to simulate position inside a hangar.

Eliminates dead spots and the need to shuffle planes.

Stay indoors in bad weather!

Repeaters are typically amplified and can cause issues when doors are open and you are near an active runway.



Use Cases

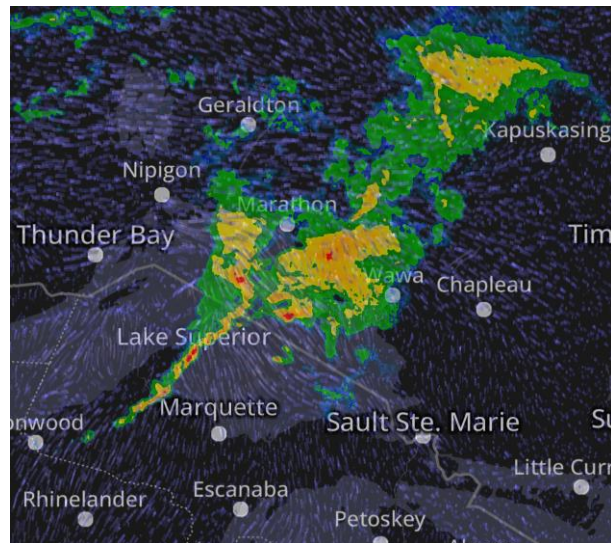
Anywhere GPS Solutions are needed

Use a static location to simulate position where there is bad weather.

Find a location where there is currently some inclement weather and determine the location coordinates.

Simulate that position to test XM Weather System

NOTE: Will require aircraft to be outdoors to allow access to XM Satellites.



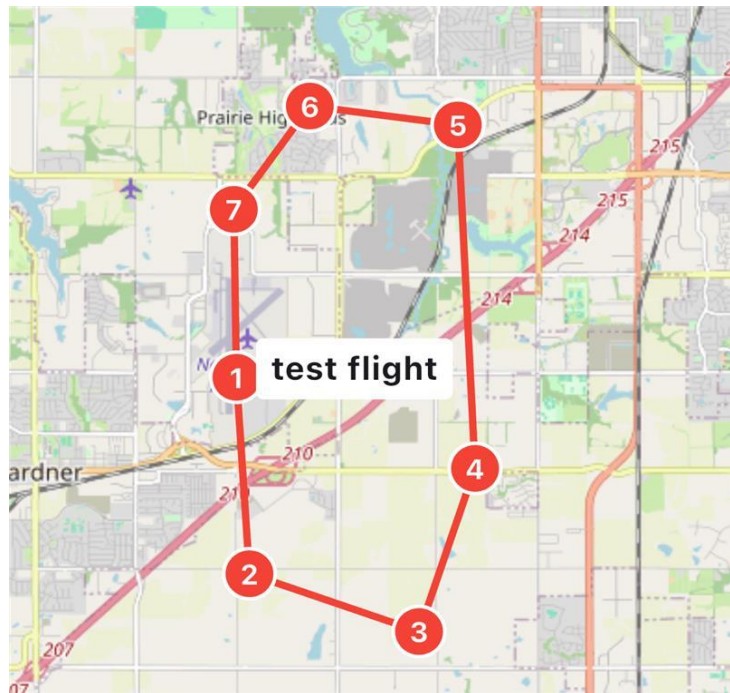
Use Cases

Anywhere GPS Solutions are needed

Use a dynamic routes to “fly” a pattern.

Fly test patterns for pilot training purposes on new cockpit installs.

Use for creating TA and RA when testing Target Generation



Troubleshooting

Getting 3D Lock

- ✓ Isolate from real satellite signals and repeaters (Turn off repeaters!)
- ✓ Apply power to GPS receiver after simulation is running
- ✓ Stay close -- geographically and in time as well (do NOT go back in time)
- ✓ Warm Start (power cycles) vs. Cold Start (clearing the GPS correlators)
- ✓ Ensure Almanac information is up to date
- ✓ Always use UTC time
- ✓ Check power levels

VIAVI GPS Solutions

GPSTG-1000 and OSPREY



VIAVI GPS Solutions

GPSSG-1000 and OSPREY

Feature	OSPREY	GPSSG-1000
Number of channels	18	12
GPS Signals	GPS L1	GPS L1, L1C, L2C, L5 Galileo E1, E5, E5a, E5b
WAAS Capable	No	Yes
Static and Dynamic support	Yes, with limited dynamics (fixed altitude/speed)	Yes, plus recording and playback of trajectories
Programmable Space Vehicle parametric and health	No	Yes
Remote Control	via Phone app	Remote Command Interface via ethernet port
Power Output Control for Sensitivity testing	Yes	Yes
Almanac download	Super simple via phone application Less than 2 minutes	Supports live satellite feed download (15 minutes) or via USB file transfer
Typical Use cases	Replacing GPS Repeaters Anything where ADS-B is needed Fly routes for pilot training after cockpit upgrade XM Weather Radar testing	Situations where more control or WAAS is required (e.g. LPV approaches)
Ease of use	Super Easy	Complexity Rating High

OSPREY GPS Simulator

Operational **S**imulated **P**ositioning for the **R**ead-time **E**valuation of **sY**stems

VIAVI's GPS Simulator provides static waypoints and dynamic routes to be simulated for avionics testing purposes.

Daily Almanac data can be seamlessly uploaded to the OSPREY, simplifying the process of ensuring a position lock for GPS Receivers.



OSPNEY Placement

Single and Dual Coupler Options

- The single coupler option can be used to test pilot and copilot GPS systems individually with the alternate GPS system disabled.
- The dual coupler option can be used to test both pilot and copilot GPS systems simultaneously.
- Direct connection to the LRU is also available as an option.



Mobile Application

So easy a caveman can do it...



Fig 1: Remote application home screen

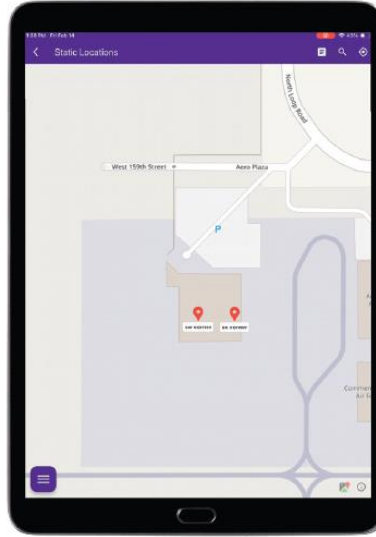


Fig 2: Remote application with static waypoints

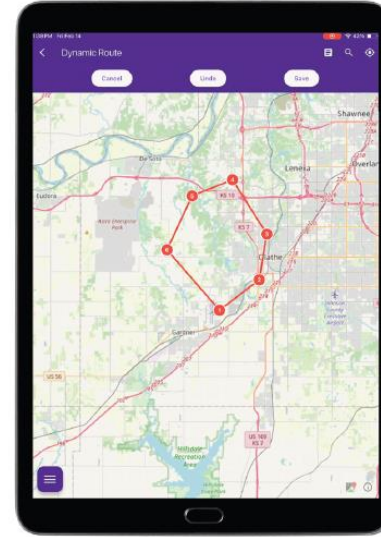


Fig 3: Remote application with dynamic waypoints

GPSG-1000 GPS/Galileo Constellation Simulator

GPSG-1000 vs OSPREY

Use the GPSG-1000 for more complex solutions:

- GPS/Galileo Portable 12-Channel Simulator
- WAAS capable
 - Required for RNAV approaches
- GPS signals simulated
 - L1, L1C, L2C, L5
- Galileo signals simulated
 - E1, E5, E5a, E5b, E6
- RAIM / Satellite Health
- Record and Playback feature
- ADS-B application option
 - Required for automated ADS-B and Latency tests when configured with the IFR6000/6015 and IFF45TS
- RCI for automated testing





VIAVI Solutions

Thank you!

GPS Overview

- Who does what now?

Launch LeoLabs Animation

Quiz

Who knew there was going to be a quiz???

Is GPS spoofing all bad?

Quiz

Who knew there was going to be a quiz???

Is GPS spoofing all bad?

